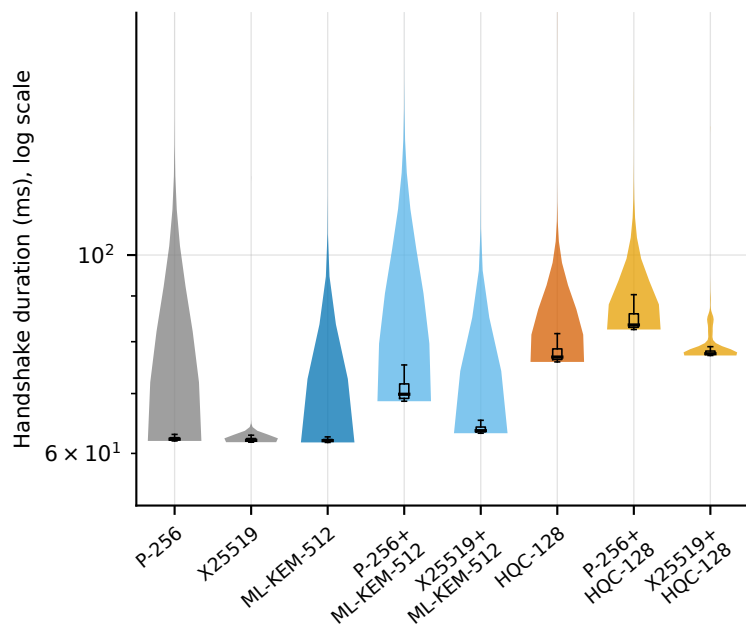


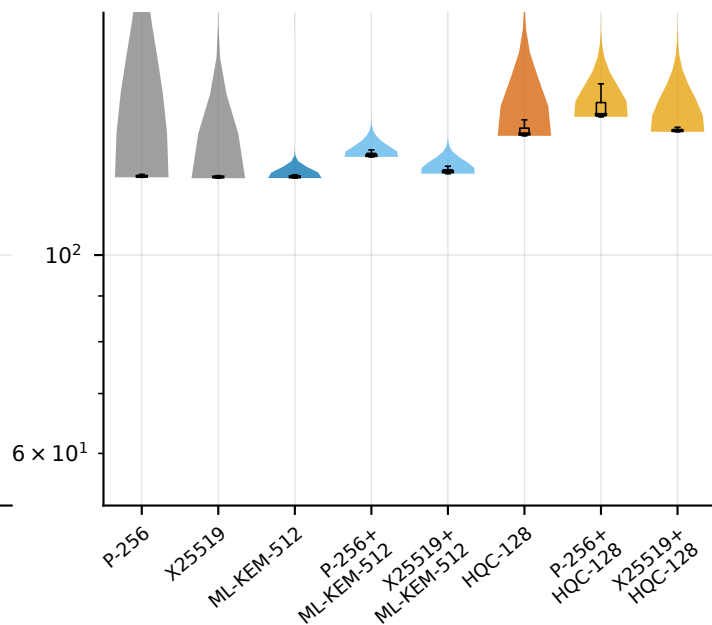
(a)

ed25519 (classical)



(b)

mldsa44 (post-quantum)



(c)

■ Classical KEX
 ■ ML-KEM (pure)
 ■ ML-KEM (hybrid)
 ■ HQC (pure)
 ■ HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	68.5	112.8	1.06k	138.4	105.7	1.58k
X25519	62.3	4.4	122.5	125.3	52.2	1.58k
ML-KEM-512	64.4	75.5	1.15k	122.8	6.9	300.8
P-256+ML-KEM-512	77.6	108.5	1.16k	130.6	6.7	309.1
X25519+ML-KEM-512	66.2	73.5	1.15k	125.0	9.3	303.3
HQC-128	80.7	43.2	622.9	142.8	39.2	1.21k
P-256+HQC-128	87.3	28.6	627.6	148.2	19.6	727.3
X25519+HQC-128	78.9	4.8	139.3	142.6	27.4	725.0